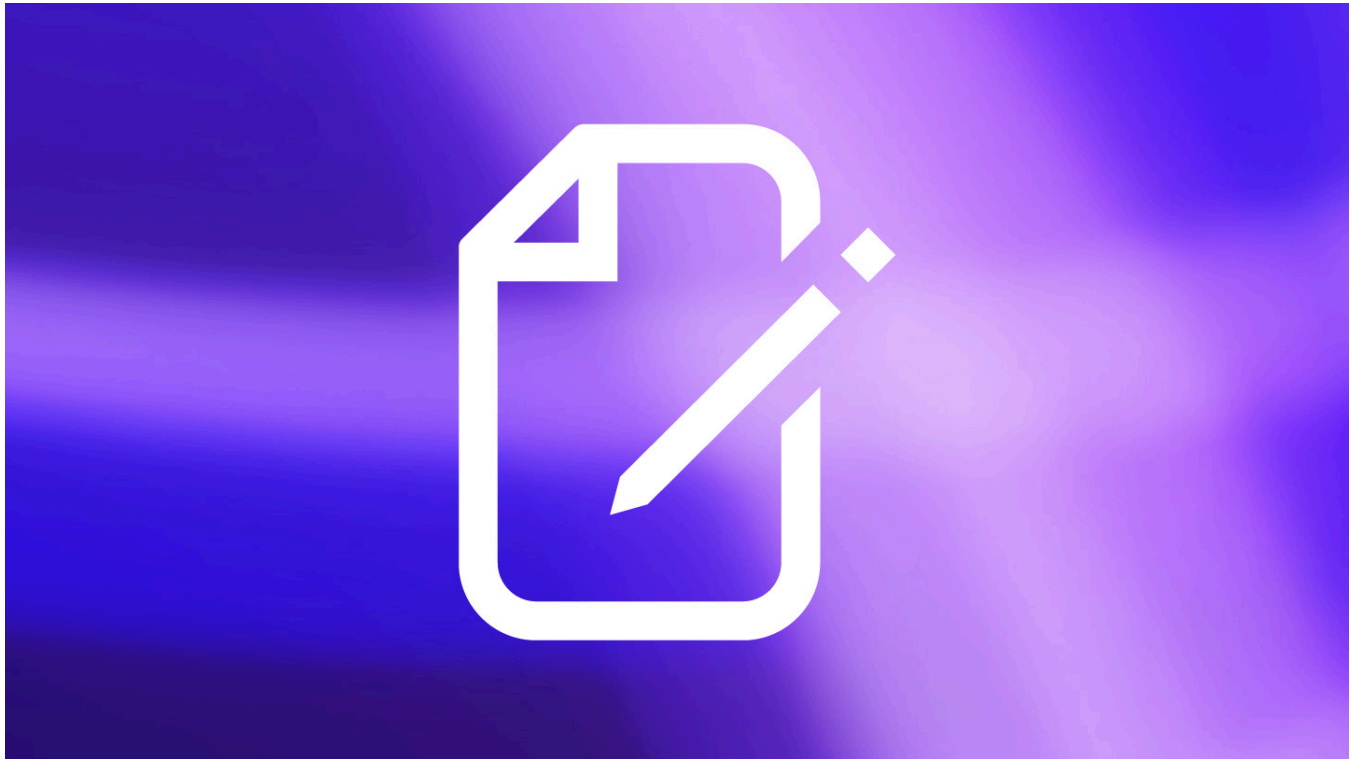
National

Operational Excellence Standard (OES) — Premise bonding requirements

HOW22918 | MODIFIED 10/26/2026



Summary

The bonding standard defines bonding requirements that all field technicians must be aware of. Additionally, the requirements of the National Electrical Code are also highlighted.

Quick Links

- At a glance
  - Required tools and hardware
  - Guidelines
- Bonding requirements
  - Approved hardware chart
- Bonding options at the dwelling or building
- Bonding attachment options — General
  - Bonding option one
  - Bonding option two
  - Bonding option three
  - Bonding option four
  - Bonding option five
  - Bonding option six
- Premises-powered broadband communications systems — Fiber to the premise
- Commercial building bonding requirements
- Bonding for mobile homes and construction trailers
- Solar power bonding
- OES document owner, contributors, and approvers
- What's changed in this article

At a glance

This article covers the OES standards for premise bonding requirements.

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Required tools and hardware

CIFA	Approved bonding hardware
306745	Clamps, bus bar, ground lug, kit.

CIFA	Approved bonding hardware
18899	24 inch. Grounding strap. Galvanized.
309537	16in, grounding strap.
18128	Galvanized. Ground strap. 6-inch. Pn# si-0958-6g. Senior industries.
18129	Galvanized. Ground strap. 9-inch. Pn# si-0958-9g. Senior industries.
18130	Galvanized. Ground strap. 12-inch. Pn# si-0958-12g. Senior industries.
18137	Ground wire. Clamp. C/vise type. #6 sol-#8 str wire range.
19597	Grounding. Clamp. Pipe. 5.8 - 1-inch. Bronze. Pn# si-5948g. Senior industries.
24657	Ground clamp. Structural for trailers. Accepts up to #6 awg. Senior industries. Pn# si-2145a.
24659	Meter ground. Clamp. Accepts up to #6 awg. Senior industries. Pn# si-2167.

CIFA	Approved bonding hardware
80584	Side lip mount. Wide lip. Meter box clamp. Grounding clamp. For #18 - #6 awg. Ul listed. Attachment stud-3/8" thread, 7/16" hex size. Ground screw 1/4-inch thread, 3/8-inch hex head.
18114	Meter bonding. Lip clamp. Pn# si-2167gc. Senior industries.
27464	Side lip mounted. Meter clamp. Accepts up to #6 awg wire. UL (Underwriters' Laboratories) listed.

Guidelines

- Questions or clarification regarding the content of this standard should be directed to the Technical Operations Leadership for the local region or market.
- This document does not substitute for any local, state, or federal code or requirements, but should be used as a supplement to these.
- **Deviations:** Practices and configurations outlined in this document are the Comcast standard and may not be modified in their implementation. Any deviations from the standards prescribed in this document shall be approved by Comcast Corporate Technical Operations.
- **Safety:** All personnel are required to always comply with Comcast’s security and safety regulations.
- **Workmanship:** All work performed will be of the highest possible workmanship. All work will adhere to Comcast’s standards and local, state, and federal codes

and requirements and be fully compliant to any stipulations made by the landlord and agent in control of the property.

Prerequisites:

- All technicians must be trained in Occupational Safety and Health Administration (OSHA), National Fire Protection Association (NFPA) 70 (National Electrical Code), and proper use of Comcast standards Personal Protective Equipment (PPE).
- All technicians must meet all state electrical licensing or certification requirements according to federal, state, and the local authority having jurisdiction where the work is being performed.
- All technicians must comply with local electrical codes, as some jurisdictions do not permit certain bonding methods.
- All technicians must be properly trained in the use of Foreign Voltage Detector (FVD) and the Digital Multi-Meter (DMM) with amp clamp for proper foreign voltage and current on coax detection.
- All field personnel, technicians, and supervisors must meet the yearly certification and recertification requirements.

Bonding requirements

Bonding and grounding shall comply with the applicable requirements of NFPA 70 (NEC) for the communications or limited-energy system installed, including requirements applicable to communications circuits, optical fiber cables, coaxial systems, and building-entry conductors and must be UL-listed.

Approved hardware chart

The below table is ordered by most preferred method to least preferred.

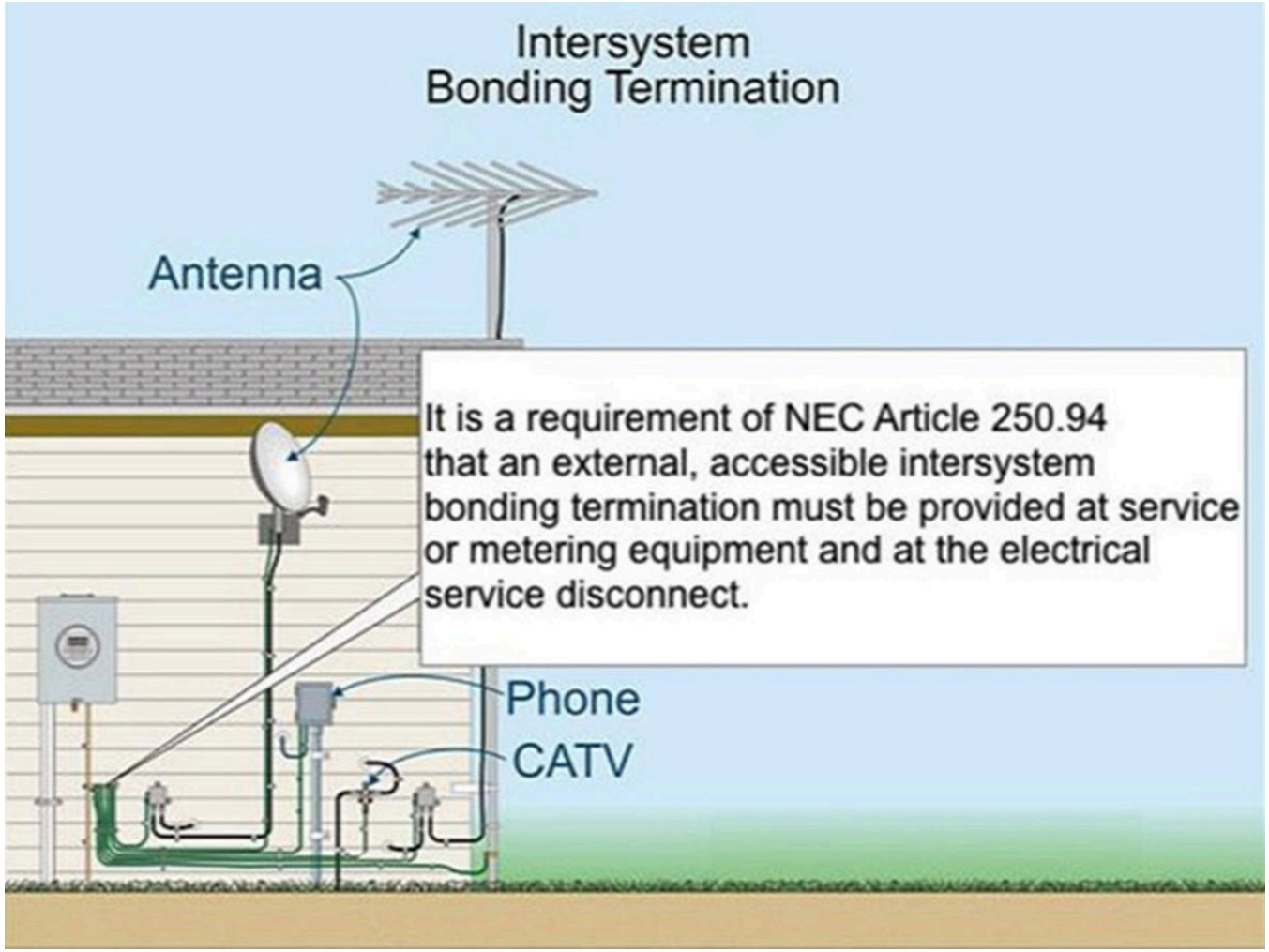
Priority	Bonding option	Approved hardware
1	Intersystem bonding bridge	No additional hardware needed
2	Copper stub or ground wire	H clamp, split bolt, single ground connector
3	Ground rod	Ground rod clamp, acorn
4	Grounded metal conduit	Galvanized strap
5	Metallic meter cabinet	Lip mount clamp, plate style connector, corner connector, bracket style clamp
6	Interior cold-water pipe	Water pipe clamp
*	Trailer frame bond	Trailer ground clamp, I beam ground clamp

An FVD and an Amp Clamp must be used to check for voltage and/ or current before beginning any work on Comcast's or the customer's network.

- If foreign voltage or current is detected, then a supervisor must be notified.
- If voltage or current exists on the customer's RF network, all work must stop, and the customer must be notified.
- No work can continue until the foreign voltage or current issue is corrected. Reference current on coax standard.

Network powered devices that are a part of residential service installations must not pass network power to the drop and must meet the requirements of Article 820 of the NEC. The provisions of Article 830 of the NEC do not apply. In MDU locations where hardline coax is installed at the premise and is used for the distribution of power for network devices, all requirements of NEC Article 830 must be implemented.

Residential and commercial coaxial drop systems shall be bonded in accordance with applicable NFPA 70 (NEC) requirements, including Article 250.94 and applicable provisions governing coaxial, communications, limited-energy, and building-entry systems. Bonding installations shall also comply with the National Electrical Safety Code (NESC) and all applicable local codes.



All coaxial cables must be bonded to the property's electrical service grounding electrode from a bonding block installed at the service distribution point.

- The bond block location should be aligned with the other utilities, located on the exterior of the premise whenever possible and in compliance with **OES Distribution Point**.
- If the utilities are located on the interior of the home or if the ideal installation location does not meet these requirements, contact your supervisor for alternate location approval.

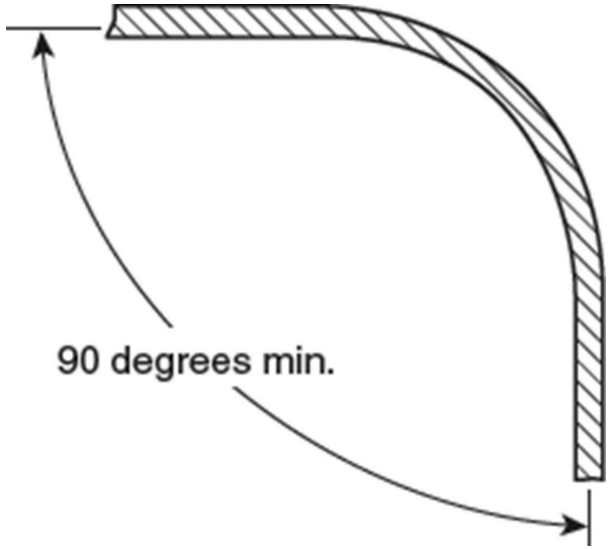
It is prohibited to bond a coaxial drop system to any natural gas piping system. This includes the ground wire and any attachments used in the practice of bonding of the natural gas piping system.

- NEC 250.52(B) not permitted for use as grounding electrodes. The following systems and materials shall not be used as grounding electrodes: (1) Metal underground gas piping systems.



The bonding conductor

- The coaxial bonding conductor shall be solid copper and shall comply with the applicable grounding and bonding requirements of NFPA 70 (NEC), including Articles 250 and 820, as applicable to the installation.
  - The bonding conductor for bonding Series-6 drop cable must be #12 AWG or larger, insulated solid copper wire.
  - The bonding conductor for bonding Series-11 / 320 cable must be #10 AWG or larger, insulated solid copper wire.
- The bonding conductor shall be routed from the bonding block to the approved bonding or grounding point by the shortest practical route, as straight as practicable, with bends and curves minimized. The bonding conductor shall not exceed 20 feet in length for residential or commercial Comcast installations. All installations shall comply with applicable grounding and bonding requirements of NFPA 70 (NEC) and applicable local codes.
- Bonding conductor (ground wire) bending requirements:
  - Sharp bends must be avoided because the sharp bend increases the impedance and may produce flash points.



  - Bonding conductors (ground wire) of all sizes must maintain a minimum bending radius of 1 ½ inches. No bend of a conductor must form an included angle of less

than 90° (per NFPA 780).
- Bonding conductors (ground wire) must be fastened/ secured at least every 18 to 24 inches if routed horizontally and every 30 to 36 inches if routed vertically with the appropriate clip.

Tagging requirements:

- Two **Do Not Remove** tags must be placed on all bonding conductors (ground wire). One at the connection point to the bond block and one at the point of its attachment to the property power grounding electrode.
  - The tags must have the date recorded in writing of when the bond was originally verified.
  - The date does not need updated unless changes are made to the bond or bonding.
  - Tag(s) must be replaced if they are unreadable.

It is prohibited for any existing bonding clamps or fasteners to be loosened or disturbed. Therefore, it is prohibited to attach the CATV bonding conductor (ground wire) under the same clamp or fastener of any existing bond clamp or fastener unless specifically designed for multiple bond attachments.





The CATV bonding conductor (ground wire) must be attached first to the property power grounding electrode with an approved fastener, then to the bonding block.

The CATV bonding conductor (ground wire) must be run from the bonding block to the premise electrical ground system without splices or any other connections.

- Only one bondblock can be attached to a CATV bonding conductor (ground wire).

The drop must always be properly bonded **before** any customer equipment is connected.

- If a drop is already bonded and the bond needs to be removed for any reason, the interior wiring must be disconnected first.

A supervisor must be notified if the bonding options listed in this standard are not available and before connecting any customer equipment to the drop system.

- The drop must be bonded before an installation can proceed.
- The customer shall be notified if a valid bond cannot be achieved.

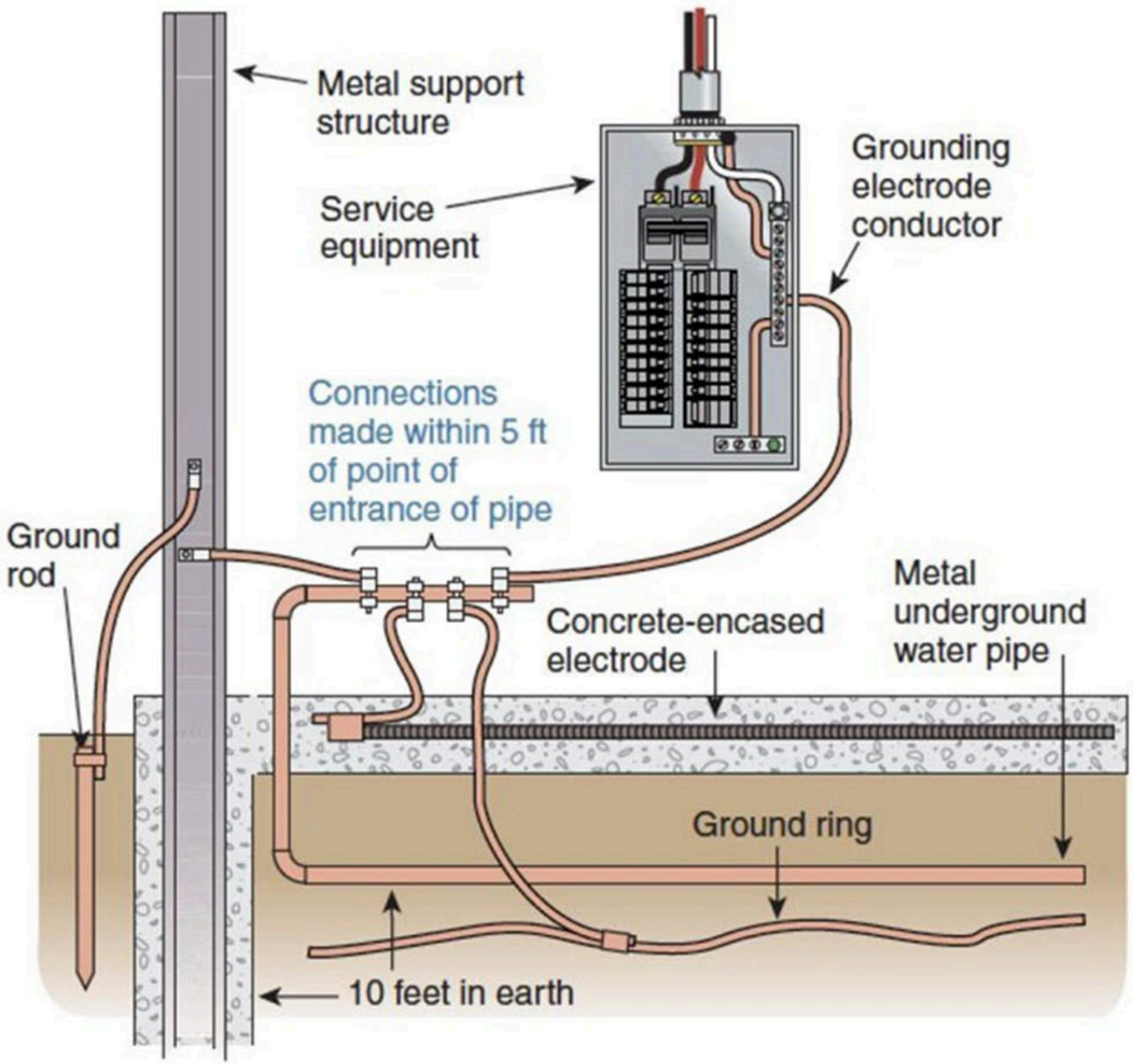
All bonds must be verified using a DMM. The resistance between the bond block and the property power grounding electrode must be less than 25 ohms.






Bonding options at the dwelling or building

The diagram depicts a typical grounding electrode system for commercial and residential applications; however, in a residential scenario, the metal support structure would be absent.



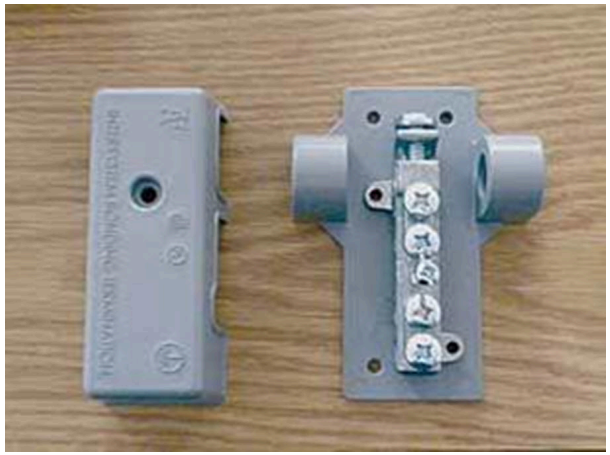
Bonding attachment options — General

All bonding options **must** be used in the order as presented based on the availability to bond. We refer to this as hierarchy.

The hierarchy is listed below in order of priority from the most preferable (bonding option one and bonding option two) to the least preferable (bonding option three through option six).

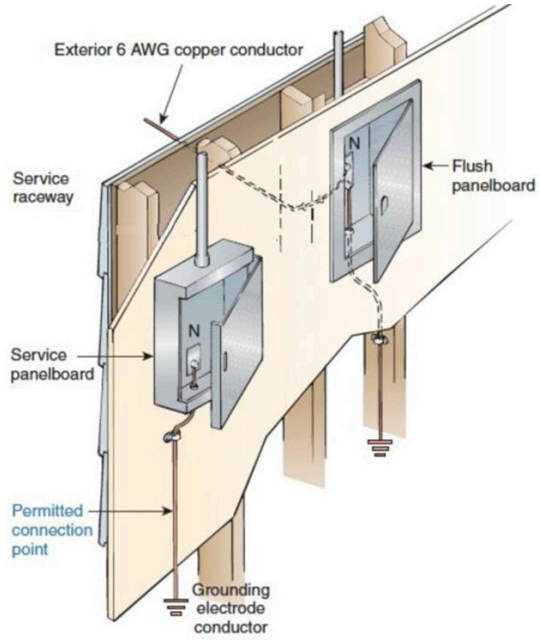
Bonding option one

The CATV bond block may be bonded to the bonding bridge that is bonded to the property power grounding electrode, if one has been made available on the outside of the property.



Bonding option two

The CATV bond block may be bonded to a #6 AWG solid copper stub bonded to the property power grounding electrode if one has been made available on the outside of the property.

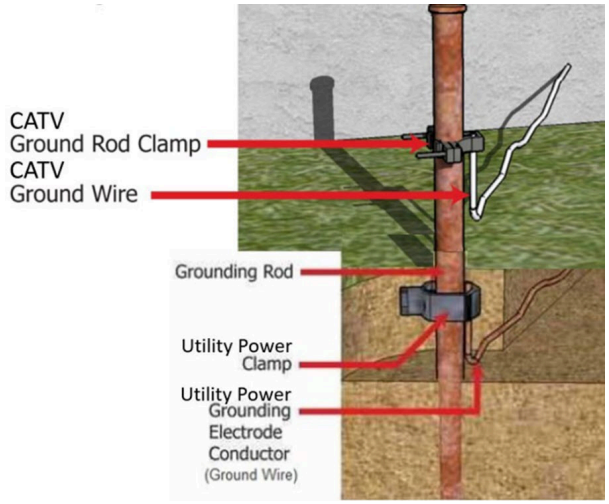


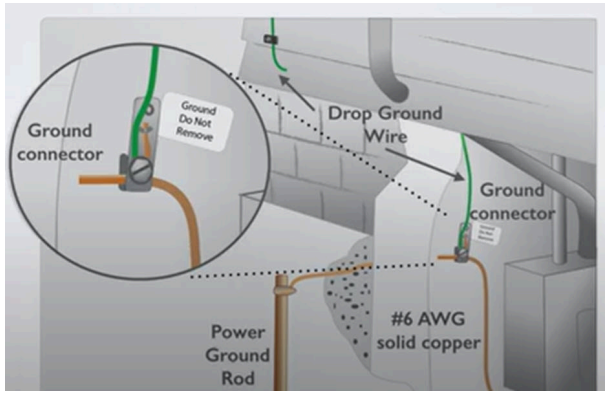
Bonding option three

The drop may be bonded directly to the main grounding electrode (i.e., ground wire or ground rod) of the electrical service.

It is not permissible to loosen or move any other attachment to the main grounding electrode (i.e., ground wire or ground rod).

The connection to the grounding electrode must be above ground; it is prohibited to connect below ground surface.



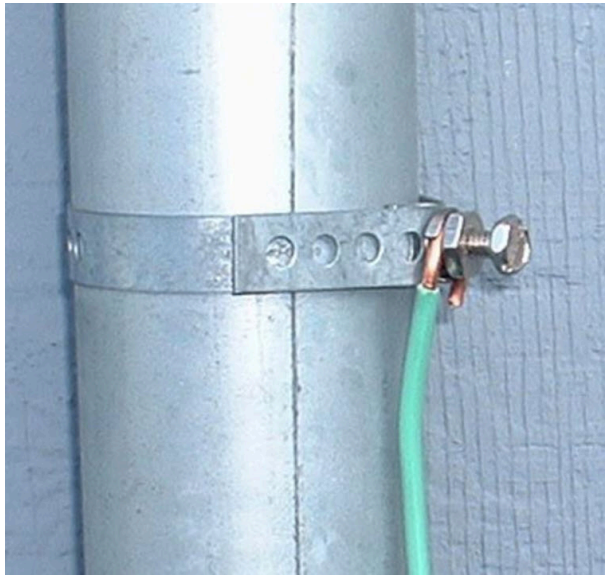


Bonding option four

The drop may be bonded to the grounded metal conduit of the electrical service mast, or the metal conduit between the utility meter enclosure and the main circuit breaker panel or sub-panels, using an appropriate and approved bonding strap.

Paint, rust, and corrosion must be removed from the conduit before attaching a bonding clamp or strap.
The metallic composition of the bonding strap and the conduit must match; a galvanized bonding strap must be used on galvanized metallic conduit.

It is prohibited to connect bonding straps to conduit servicing branch circuits from the electrical service panel.



Bonding option five

The drop may be bonded to the metallic meter cabinet using an appropriate and approved clamp. It is prohibited for the bonding attachment to be made to the electrical equipment door or other movable components of the metallic meter cabinet or any associated screws.

Electrical equipment boxes or cabinets must never be opened for any reason. Only a licensed electrician should access the inside of any electrical cabinet.



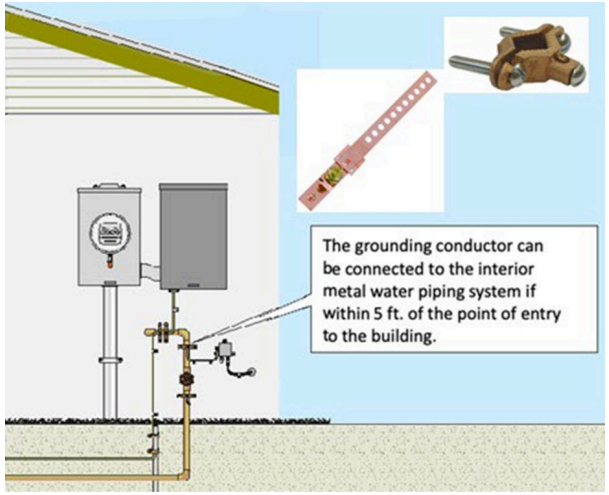
Bonding option six

The drop may be bonded to the grounded interior metal cold water piping system using an appropriate and approved pipe ground clamp. Bonding to metal cold water pipe is acceptable only when the bond can be made within five feet of where the metal underground cold-water pipe enters the property, and the metal underground cold-water pipe has been in direct contact with the ground for 10 feet or more before the point of entrance to the property. The below conditions must also be met.

- If a water meter is present, the water pipe ground clamp must be installed between the point of entry of the water pipe and the water meter.
- If a bonding clamp cannot be installed between the point of entry and the water meter, then a bond clamp can be installed on the output side of the cold-water meter only when a #6 AWG grounding jumper present and is jumping the cold-water meter.
- This option must only be used when the electrical system is also bonded to the cold-water pipe.
- A DMM must be used to verify that the interior metal water piping system is properly grounded (i.e., with a verified, measured resistance of less than 25 ohms, and verified

continuity between the water pipe bonding connection point and the electrical service grounding electrode).

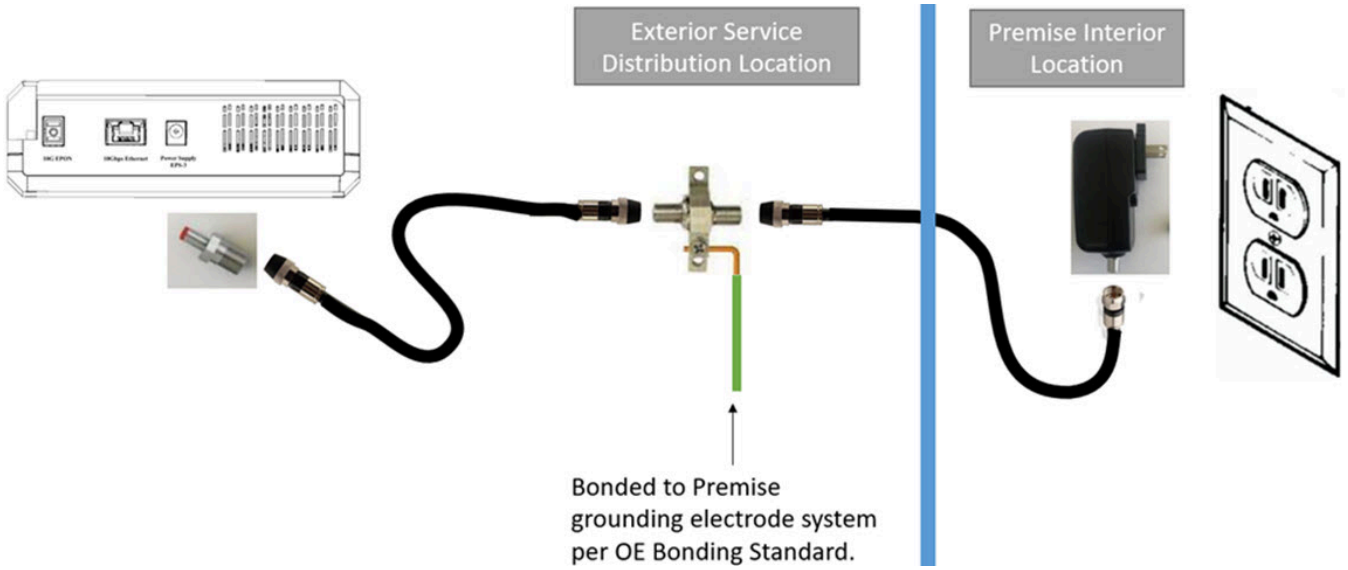
- If these conditions are not met, notify your supervisor and customer that a valid bonding option is not available.
- It is **prohibited** to bond the drop to an external cold-water faucet.



Premises-powered broadband communications systems — Fiber to the premise

- All bonding requirements of this standard apply to all fiber RFOG installations.
- A bond block must be used, bonding directly to the RFOG ONU is not allowed.
  - For EPON installations, where coaxial cable is being used the cable must be bonded. These situations are explained below.
- When using coaxial cable to feed power from the EPON XPS12 to the ONU, the coaxial line must be bonded to the premises grounding electrode system utilizing a bond block.
  - This cannot be a MoCA bond block as these do not pass power
- When using coaxial cable to supply video services through MoCA to XiDs the coaxial lines must be bonded to the premises grounding electrode system utilizing a bond

block.
    - These lines must be bonded regardless of whether they exit the premise or not.
- Refer to **OES EPON requirements** for examples of both bonding situations.
- All fiber optic installations must meet the requirements of NEC Article 770 as applicable.
- All fiber optic installations must meet the grounding and bonding requirements as listed in NEC 770.93: Grounding, bonding, or interruption of non–current-carrying metallic members of optical fiber cables.
- All fiber optic installations must meet the requirements of NEC 770.100: Entrance cable bonding and grounding.



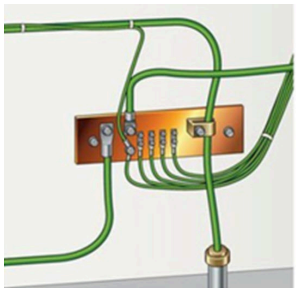
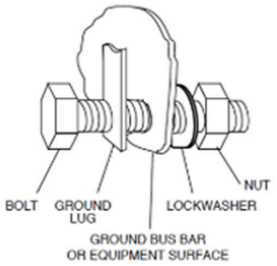
Commercial building bonding requirements

- All bonding requirements of this standard apply to stand-alone commercial building applications.
- For multi-dwelling and commercial applications where the customer service is supplied directly from a tap, the tap must be properly grounded and bonded per National Engineering and Construction Specifications.

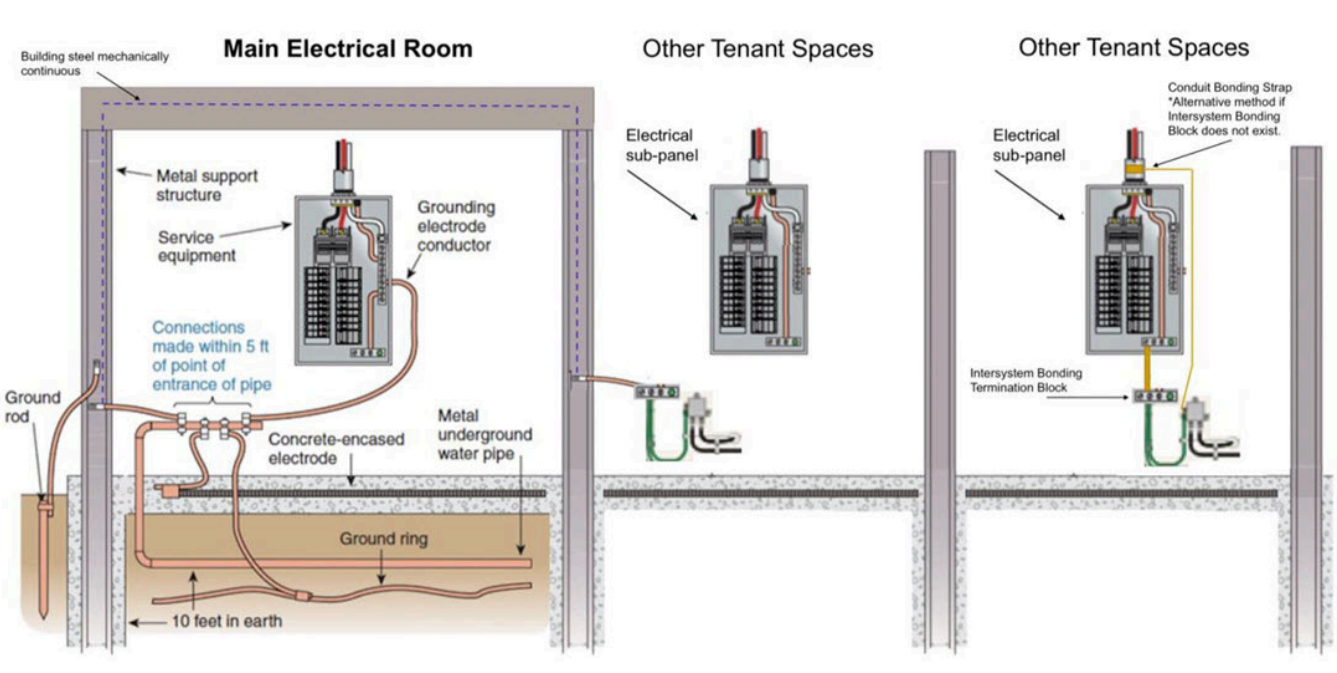
- A properly bonded tap can be identified by a #6 ground wire running from the tap or mounting plate, to which the tap is connected, to the property power grounding electrode system.




- In multi-story buildings or strip-mall applications when the tenant spaces are electrically fed from an electrical sub-panel within the tenant space, the bonding methods as listed below and depicted below must be adhered to if the cable drop is installed to the tenant space without passing through the main electrical room or following the main point of entry.
  - Intersystem bonding termination block or bus bar connected to the property's electrical service grounding electrode system.

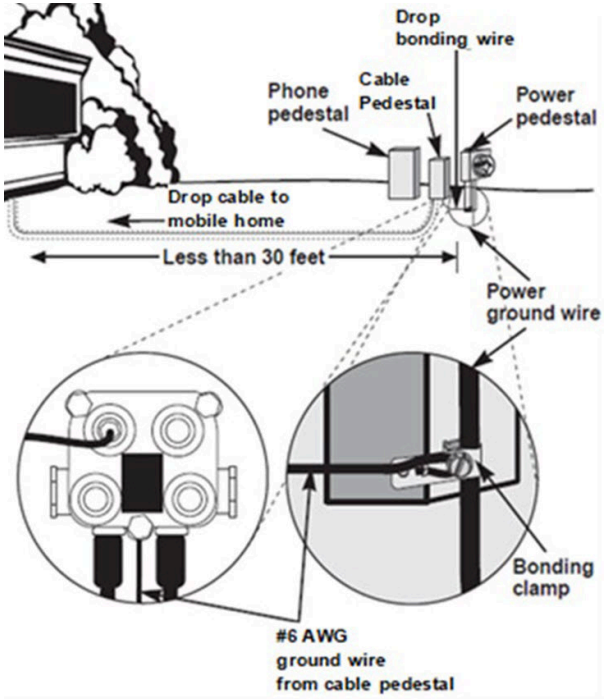
  - Bonding to building steel, which must be mechanically continuous and bonded to the main electrical room buildings' steel grounding. Bonding to building steel that is not mechanically continuous is prohibited.
  - Electrical sub-panel metal conduit grounding strap (see Bonding Option 5).



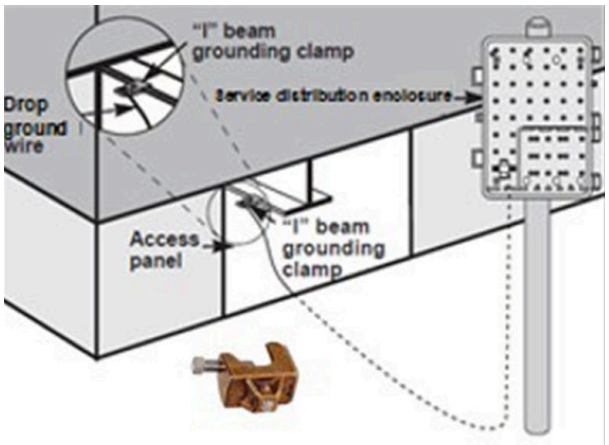
Bonding for mobile homes and construction trailers

- Grounding and bonding of communications service installations at mobile homes and construction trailers shall comply with the applicable requirements of NFPA 70 (NEC). Coaxial installations shall comply with applicable requirements for coaxial systems, including Articles 250 and 820, while fiber-to-the-premises installations shall comply with applicable requirements governing optical fiber cables and conductive components. All installations shall also comply with applicable NESC and local code requirements.
- If the power service equipment is within 30 feet of the structure, follow the bonding guidance provided above in **Bonding options at the dwelling or building**.
- If the tap pedestal location is co-located with a power pedestal and both are within 30 feet of the mobile home, drop bonding to the tap is permissible (though not a preferred option) provided the tap is properly grounded and bonded per National

Engineering and Construction Specifications.



- If the power service equipment is more than 30 feet from the structure or if the structures electrical service is provided by a cord and plug, an approved I beam clamp connected to the metal frame with a ground wire not smaller than 12 AWG connected to the bonding block is permissible.



Solar power bonding

The presence of a solar photovoltaic (PV) system does not eliminate or modify the requirement to properly bond the coaxial cable drop. PV systems shall comply with applicable NEC Articles 690 and 250. The coaxial cable drop shall be connected to an approved premises bonding or grounding point in accordance with NEC 250.94 and other applicable NEC requirements. Do not use PV equipment, racking, or associated components as a coaxial bonding point unless specifically permitted for that purpose by the applicable NEC requirements.

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What's changed in this article

Date	Summary of change
October 26, 2026	Updated to reflect NFPA 70 (NEC) updates.
November 24, 2025	Updated prerequisites in Guidelines section.
September 15, 2025	Added to Celestial.
March 27, 2024	Conducted annual review and updated: • Added current over coax as a referenced document. • Added amp clamp as a recommended tool. • Added reference to current on coax requirements. • Removed no longer valid ground wire half the distance of the shortest RF outlet. • Added requirements for working on existing bonded drop. • Changed order of bonding options four and five based on NEC updates.
June 23, 2022	Conducted annual review and updated.

↑ Back To Top